CO3-AT1

1)
```
def merge_sort(arr):
    if len(arr) > 1:
        mid = len(arr) // 2
        left = arr[:mid]
        right = arr[mid:]
        merge_sort(left)
        merge_sort(right)
        i = j = k = 0
        while i < len(left) and j < len(right):
            if left[i] < right[j]:
                arr[k] = left[i]
                i += 1
            else:
                arr[k] = right[j]
                j += 1
            k += 1
        while i < len(left):
            arr[k] = left[i]
            i += 1
            k += 1
        while j < len(right):
            arr[k] = right[j]
            j += 1
            k += 1
arr = [38, 27, 43, 3, 9, 82, 10]
merge_sort(arr)
print("Sorted array:", arr)
```

OUTPUT:

Sorted array: [3, 9, 10, 27, 38, 43, 82]

2) def binary_search(arr, x):

```
    left = 0
    right = len(arr) - 1
    count = 0
    while left <= right:
        count += 1
        mid = (left + right)
        if arr[mid] == x:
            return mid, count
        elif arr[mid] < x:
            left = mid + 1
        else:
            right = mid – 1
   return -1, count
arr = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
index, iterations = binary_search(arr, 7)
print("Element found at index:", index)
print("Iterations:", iterations)
```

OUTPUT:

Element found at index: 6

Iterations: 4